

CortexX AI System

An AI-Native Autonomous Engineering System

Built with Next.js & FastAPI

Project Overview

- CortexX is a next-generation Multi-Agent AI system.
- Frontend: Stunning Next.js Glassmorphism UI.
- Backend: High-performance Python FastAPI server.
- LLMs: Powered by Google's Gemini advanced models.
- Two core modes: General Chat Copilot and Project Builder.

General Chat Copilot

- Intelligent routing to specialized AI sub-agents.
- Code Intel Agent: Analyzes architecture & refactors code.
- Debug Agent: Rapid log parsing & root-cause identification.
- DevOps Agent: CI/CD pipeline monitoring & troubleshooting.
- Knowledge Agent: Semantic search across the codebase.

Autonomous Project Builder

- Gives the AI physical agency over your local machine.
- Provides absolute workspace paths for code generation.
- LangChain ReAct loop executes tasks iteratively.
- Tool integrations: Read files, perform surgical edits.
- Executes terminal commands to install & run apps.

Conclusion

- Bridges the gap between passive chat and active engineering.
- Acts as an autonomous digital teammate, not just a generator.
- Secure local deployment with Cloudflare tunneling.
- Fully capable of building & maintaining complex systems.
- Revolutionizing how developers interact with codebases.